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Mechanism of ring beam influence on spatter and surface collapse in full penetration laser welding of aluminum alloys

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Highlights

- The first attempt was made to apply the ring beam to improve the spatter and weld surface collapse problems.
- The ring beam prevents the generation of liquid columns and spatters by reducing melt vapor shearing on the melt.
- The upward flow, dominated by the ring beam at the top of the molten pool, can effectively alleviate weld surface collapse.
- The improvement mechanism of the ring beam on spatter and surface collapse under full penetration conditions was revealed.

Abstract

Adjustable-ring-mode laser welding (ARMLW) exhibits significant potential for enhancing the surface quality of aluminum alloy welds. However, its application in improving spatter and weld surface collapse during full penetration laser welding (FPLW) of medium-thick plates remains unclear. This study combined experimental and numerical simulation methods to first examine the quantitative impacts of the ring beam on the weld surface roughness and surface collapse height. Subsequently, a 3D transient CFD model was established to investigate the impact of the ring beam on the dynamic behavior of the keyhole and the flow characteristics of different regions and sections of the molten pool, thereby elucidating the mechanism by which the ring beam improves spatter and weld surface collapse. The results demonstrated that the ring beam can significantly enhance the surface roughness of the weld. The surface collapse height was reduced by 14.07% with a ring power of 1500W. Compared with a ring power of 0W, the addition of the ring beam resulted in a stable, regular through-keyhole by enlarging the keyhole entrance and alleviating the oscillation of the keyhole. The more uniform energy distribution on the keyhole wall enabled the melt to exhibit multiple, regular flows over a long distance. The lower level and upward flow velocity at the top of the molten pool effectively mitigated the liquid column and spatter generated by the upward flow. Moreover, the lower downward flow velocity in the middle and bottom and the upward flow velocity dominated by the ring beam at the top of the molten pool effectively suppressed the fall of the melt. It diminished the weld surface's collapse height.

Keywords

Adjustable-ring-mode; Full penetration laser welding; Weld surface collapse; Keyhole behaviors; Molten pool flow characteristics

CRediT authorship contribution statement

Declaration of competing interest

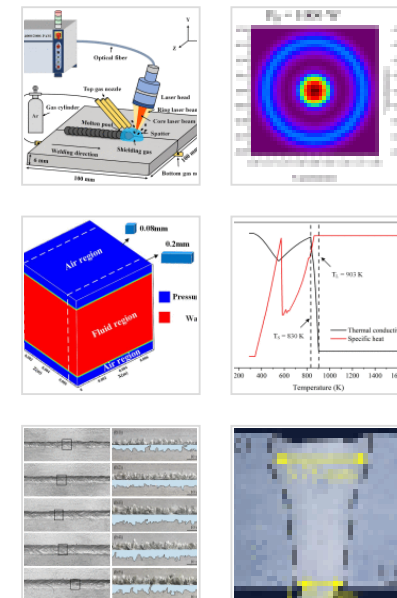
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1. Introduction

Laser welding technology is one of the most reliable ways for joining aluminum alloys [1]. With the development of laser equipment, the intense penetration of high-power lasers enables single-pass FPLW of aluminum alloys. Compared to partial penetration welding, FPLW can provide a continuous and complete force flow transmission path for the structure, ensuring that the effective load-bearing area at the joint is equal to or close to the cross-sectional area of the base material, thereby providing excellent performance assurance [2]. However, in the FPLW process of medium-thick aluminum alloy plates, the high energy density input leads to highly transient keyhole and molten pool behavior, resulting in more complex oscillation and flow patterns. Spatter resulting from an unstable welding process [3], weld surface collapse, and root hump due to insufficient molten pool support [4,5], as well as internal porosity [6,7], thereby limiting the application of FPLW. These defects must be addressed immediately by proposing corresponding suppression methods. It is known that the aforementioned defect problems are intimately connected to the dynamic behavior of the keyhole and the heat flux distribution within the molten pool. Therefore, precisely managing them to produce superior welding joints is one of the more reliable research directions at present.

To address the above problems, oscillation laser welding (OLW) [8,9], back protection gas [10], external energy field assistance [11], and sub-atmospheric pressure environment [12] have been commonly employed. Cai et al. [13] conducted an in-depth study on the effects of oscillation parameters on the energy distribution, weld formation, and internal defects of OLW of aluminum alloy. The experimental results showed that OLW improved weld surface consistency while eliminating surface spatter, undercut, and collapse problems. Mu et al. [14] applied a longitudinal alternating magnetic field to laser welding of aluminum alloys. They examined the quantitative impact of the magnetic field on weld formation, porosity defects, and microstructure in detail. According to the study, the alternating magnetic field formed a more uniform and finer fish-scale pattern on the weld surface and bottom, alleviating surface fluctuations while significantly reducing porosity. Jiang et al. [15] compared FPLW under atmospheric and sub-atmospheric

Table 3. Process parameters used for las...

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pressures, examining the weld surface formation and cross-sectional morphology under various ambient pressures. The results demonstrated that penetration increased dramatically with decreasing ambient pressure, and high-quality FPLW joints with a smooth surface appearance could be obtained at pressures lower than 0.1 kPa. The laser intensity was only half that at standard atmospheric pressure. All of the aforementioned studies provide somewhat efficient methods, but they are also constrained by the challenge of matching intricate process parameters and establishing vacuum and electromagnetic environments.

In recent years, ARMLW technology based on beam shaping has demonstrated unique advantages in improving welding quality and molten pool stability due to its higher adjustability and flexibility [16]. By regulating the power distribution of the ring beam, it can effectively control the flow of the molten pool. Current research primarily studies the effect of different ring powers and power ratios on surface forming [17,18], internal defects [19,20], microstructure and performance [21] through process experiments combined with characterization tests, and to a certain extent establishes a quantitative relationship for the effects of ring beam processes. However, when it comes to studying the physical mechanism of the ARMLW, existing experimental observation methods are complex and do not accurately capture the transient behavior of the molten pool. Numerical simulation, as a viable complementary tool, can provide a more thorough understanding of the fundamental physics underlying the welding process. Hu et al. [22] employed a hybrid heat source to compare the oscillation of the keyhole and the flow of the molten pool between adjustable-ring-mode full penetration laser welding (ARMFPLW) and FPLW. They discovered that the ring beam may prolong the oscillation period and reduced the oscillation amplitude of the keyhole, thereby significantly minimizing porosity defects. Qi et al. [23] noted that superimposing a coaxial ring beam can efficiently balance the damaging pressure of melt backflow and eddy current on the keyhole, resulting in a welding process with nearly no droplet splashing and a weld surface with almost no undercut, collapse, and hump. Wang et al. [24] constructed a dynamic heat-flux coupling model. They simulated that the wider keyhole entrance and

improved keyhole stability under the ring beam are the main reasons for reducing spatter and achieving a smooth weld surface. Li et al. [25] explained the process of spatter reduction using the ring beam, noting that the ring beam may suppress spatter by preheating the leading edge of the molten pool, thereby stabilizing the laser absorption rate and reducing the eruption of the vapor plume. At the same time, they also determined the standard process parameters for achieving spatter-free results. In addition, they found that the ring beam can simultaneously decrease the pores generated at the gas-liquid and liquid-solid interfaces [26]. They suggested using the smallest penetration-to-spot diameter ratio (PDR) as a criterion for selecting the ring power ratio [27]. Yang et al. [28] found that the ring beam contributes to expanding the keyhole diameter and inclination angle of the back wall, thereby increasing the keyhole wall's ability to withstand the melt effect while reducing the keyhole wall's radial pressure gradient and fluid dynamic pressure, which further reduces the porosity.

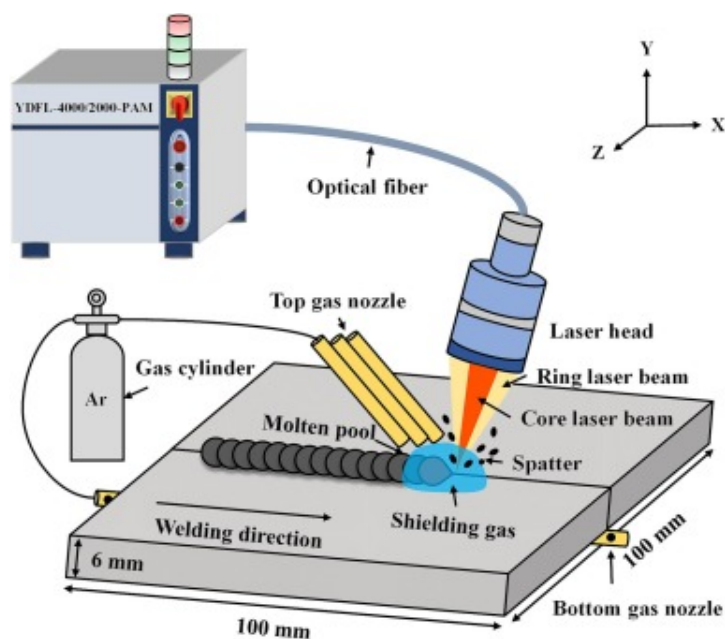
Our understanding of the ring beam's application to reduce weld surface spatter and internal porosity has increased as a result of the aforementioned investigations. However, the above research primarily focused on revealing the improvement mechanism under partial penetration process conditions, neglecting the surface spatter and weld surface collapse under FPLW. For FPLW of medium-thick plates, due to the lack of support at the weld boot and higher energy input, the dynamic behavior of the keyhole and the metal flow characteristics in the depth direction exhibit higher complexity and instability. As a result, it remains unclear how to improve the surface spatter and weld surface collapse of medium-thick plate aluminum alloys under FPLW using the ring beam.

In this paper, the effects of the ring beam on spatter and weld surface collapse during FPLW of medium-thick aluminum alloy plate are examined through both experimental and numerical investigations for the first time. The quantitative influence of the ring beam on weld roughness and surface collapse height was experimentally studied. A verified 3D transient CFD model was used to simulate the keyhole's dynamic behavior and heat flux pattern in the molten pool under the FPLW. The formation of surface spatter and weld surface collapse under FPLW, as well as the influence mechanism of the ring

beam on them, was discussed in detail. This work innovatively verified the feasibility of using the ring beam to improve spatter and weld surface collapse in FPLW, clarifying the mechanism by which spatter and surface collapse were improved. This finding will help promote the development of ring beam lasers for medium-thick plate welding applications.

2. Experimental procedures

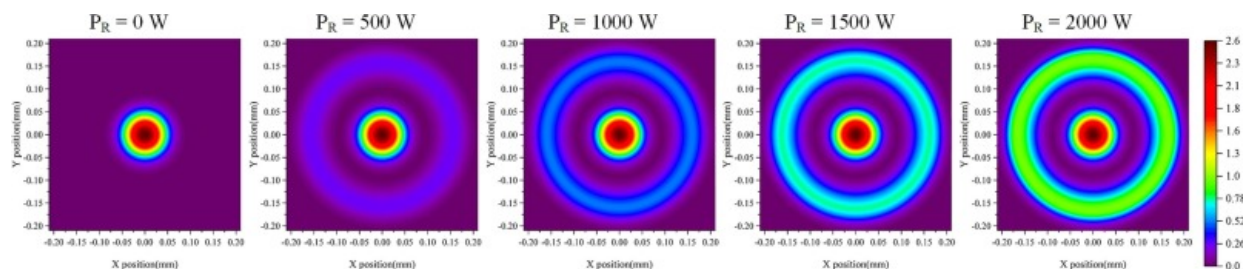
The laser welding experimental device is illustrated in Fig. 1. An ARM continuous fiber laser (YDFL-4000/2000, FEIBO, China) was selected to provide the energy input, with a total input power of 6kW. The power of core and ring beams could be modulated individually. The collimator and lens concentrate the laser beam, which has a wavelength of $1080 \pm 10 \text{ nm}$. At the focal plane, the core and ring spot diameters are 0.14mm and 0.42mm, respectively. The laser energy density distributions of various powers are depicted in Fig. 2. P_R stands for ring power.



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Fig. 1. Experimental device of laser welding.



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Fig. 2. Laser energy density distribution for different ring powers.

The laser welding experiment was conducted on 5A06 aluminum alloy plates with dimensions of 100mm×100mm×6mm. The chemical composition and mechanical properties of the base material are presented in [Table 1](#) and [Table 2](#), respectively. To eliminate the impact of surface oil and oxide coating prior to welding, the surface was polished along the welding direction and cleaned with anhydrous ethanol solution. The process parameters used in the welding experiment were based on previous process explorations. The parameter results, which show apparent surface collapse under full penetration conditions, are displayed in [Table 3](#). Since this paper exclusively investigates the influence of ring power, the following parameters are fixed: 20L/min argon gas flow, 20mm/s welding speed, and 0mm defocus. The OLYMPUS LEXT OLS5000 confocal laser scanning microscope was used to observe weld surfaces and quantify surface roughness. The measurement length was approximately 5 mm along the weld direction, and the measurement width included the entire weld width and a portion of the substrate area. Using the substrate surface as a reference plane, the surface height distribution was extracted at three locations along both the weld direction and perpendicular to it. The average maximum surface height difference in different directions quantified the

uniformity and smoothness of the weld surface. The specimens were cut by electrosark machining and inlaid with resin. After being polished to a mirror finish with 80# - 5000# diamond sand, the cross-section was etched using Keller's reagent ($V(HF): V(HNO_3): V(HCl): V(H_2O) = 1:2.5:1.5:95$). Finally, the cross-sectional morphology of the weld was observed using a KEYENCE VK-X1000 three-dimensional stereomicroscope. The cross-sections of the weld at three different locations were selected, and the average morphological size was calculated to measure the weld size.

Table 1. The chemical components of 5A06 aluminum alloy.

Element	Mg	Mn	Si	Zn	Zr	Fe	Al
wt%	6.2	0.69	0.21	0.19	0.2	0.14	the rest

Table 2. The mechanical properties of 5A06 aluminum alloy.

Tensile strength (MPa)	Yield strength (MPa)	Elongation (%)
352.5±7.2	232.9±20.4	38±1.6

Table 3. Process parameters used for laser welding tests.

Case	Core Power (P_C/W)	Ring Power (P_R/W)	Ring Power Ratio (%)	Welding speed (m/s)
P_{C4}	4000	0	0	0.02
$P_{C4}P_{R0.5}$	4000	500	11.1	0.02
$P_{C4}P_{R1}$	4000	1000	20	0.02
$P_{C4}P_{R1.5}$	4000	1500	27.3	0.02

Case	Core Power (P_C/W)	Ring Power (P_R/W)	Ring Power Ratio (%)	Welding speed (m/s)
$P_{C4}P_{R2}$	4000	2000	33.3	0.02

3. Mathematical model

The ARMLW process involves highly complex transient physical behaviors and interactions. Therefore, before modeling, the model needs to be simplified to ignore minor influencing factors. The model is based on the following basic assumptions:

- (1) The melt metal is considered to be a layered, incompressible Newtonian fluid;
- (2) The material is isotropic and homogeneous, with a density that does not change with temperature, and the Boussinesq approximation is used to consider thermal buoyancy;
- (3) The effect of the shielding gas on the melt is ignored, whether as a force or a cooling effect.
- (4) The interaction between the plasma and the laser beam is ignored, and the thermal effect of the vapor plume on the surface of the molten pool is indirectly considered.

3.1. Laser heat source model

The ARM laser beam is composed of a coaxial superposition of core and ring beams. In the one-dimensional direction, there are prominent energy density peaks in the core and the centers of the ring bands on both sides, and both the core and ring beams can be

viewed as Gaussian distributions, according to the following equation [29]:

$$I_c(r) = \frac{3\eta_c Q_c}{\pi R_{zc}^2} \exp\left(-\frac{3r^2}{R_{zc}^2}\right) (r^2 \leq R_{zc}^2) \quad (1)$$

$$I_r(r) = \frac{3\eta_r Q_r}{\pi R_{zh}^2} \exp\left(-\frac{3(r-R_{zr1})^2}{R_{zh}^2}\right) (R_{zr1}^2 \leq r^2 \leq R_{zr2}^2) \quad (2)$$

Where I_c and I_r are the laser energy densities of the core and ring beams, respectively. η_c and η_r , as well as Q_c and Q_r , are their corresponding energy utilization efficiency and input power, respectively. R_{zc} denotes the spot size of the core beam. R_{zr1} , R_{zr2} , and R_{zh} represent the spot size and half bandwidth of the ring beam, respectively.

The lasers at the focal plane were subdivided into 96×96 sub-beams to improve the modeling accuracy of the beam-material interaction. The ray tracing technique [30] was utilized to account for numerous beam reflections, and the Fresnel equation [31] was employed to represent the laser energy absorption rate at the incident/reflected position of the beam within the keyhole:

$$\alpha_{Fr}(\theta) = 1 - \frac{1}{2} \left(\frac{1+(1-\varepsilon \cos \theta)^2}{1+(1+\varepsilon \cos \theta)^2} + \frac{\varepsilon^2 - 2\varepsilon \cos \theta + 2(\cos \theta)^2}{\varepsilon^2 + 2\varepsilon \cos \theta + 2(\cos \theta)^2} \right) \quad (3)$$

Where ε is a constant linked to the material and laser parameters, and θ is the angle between the normal of the keyhole wall and the incident light.

3.2. Governing equations

In this study, a three-dimensional multiphase incompressible laminar flow model was established, and the time-space distributions of temperature, velocity, and pressure in the molten pool were determined by solving the following governing equations. The conservation equations for mass, momentum, and energy are as follows:

$$\nabla \cdot \vec{u} = 0 \quad (4)$$

$$\rho \left(\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} \right) = -\nabla P + \mu \nabla^2 \vec{u} - \rho K \vec{u} + \rho \vec{g} + \vec{F} \quad (5)$$

$$\frac{\partial(\rho H)}{\partial t} + \nabla \cdot (\rho H \vec{u}) = \nabla \cdot (\lambda \nabla T) + S_h \quad (6)$$

Where $\vec{u}$ is velocity and $\vec{g}$ is gravity, ρ represents the density, P denotes the pressure, μ is the viscosity, K indicates the Carman-Kozeny coefficient, λ is thermal conductivity, H is enthalpy, $\vec{F}$ and S_h denote the momentum and energy source terms, respectively.

Coupled Level-Set and Volume of Fluid (CLSVOF) algorithm simultaneously compensates for the lack of accuracy of the VOF algorithm in dealing with non-smooth interfaces and the mass non-conservation problem caused by the LS algorithm's larger numerical dissipation, allowing it to describe the topological evolution of complex interfaces better. As a result, it was employed in this research to record dynamic changes in the gas-liquid interface. In this study, the VOF function was mainly constructed based on the LS function in the initialization stage, which is represented as follows [32]:

$$F = \frac{1}{2}(1 + \tanh(h_s \phi)) \quad (7)$$

F , ϕ and h_s are the VOF function, LS function and interface thickness, respectively.

$$h_s = 0.85 dx.$$

Finally, the precise position of the free surface was determined by solving the VOF convection equation:

$$\frac{\partial F}{\partial t} + \vec{u} \cdot \nabla F = 0 \quad (8)$$

3.3. The major driving forces

The principal driving forces on the free surface included recoil pressure [33], buoyancy [34], surface tension [35], and vapor force [36,37], as shown in Eqs. (9), (10), (11). The vapor force was mainly divided into the shear stress τ_{vap} , and stagnation pressure P_{sta} of

the metal vapor.

$$F = ABT_s^{-1/2} \exp(-ML_v/(N_a k_b T_l)) + \rho \beta (T - T_{ref}) \vec{g} + \sigma \kappa \delta(\phi) \mathbf{n} + F_v \quad (9)$$

$$\tau_{vap} = \frac{8\rho_g(v_g \sin \theta)^2}{R_e} \quad (10)$$

$$P_{sta} = \frac{\rho_g(v_g \cos \theta)^2}{2} \quad (11)$$

A is a correction factor relating to material qualities and environmental pressures, usually 0.55, B is the evaporation constant, T_l denotes the surface temperature of the liquid phase, M , L_v , N_a , and k_b are molar mass, evaporation latent heat, Avogadro constant, and Boltzmann constant. β is the thermal expansion. T_{ref} represents the reference temperature of 300K. ρ_g denotes the vapor density, and R_e represents Reynolds number. σ indicates the surface tension value, and κ is the mean curvature of the local interface. $d\sigma/dT$ represents the gradient of tension and $\mathbf{n}$ indicates the unit normal vector of the interface.

3.4. Boundary conditions

The thermal boundary conditions of the free surface are presented in Eq. (12), where the right side of the equation represents the laser heat input, the heat flux of the vapor plume, convection, radiation, and evaporation heat loss [38]. This paper drew on the method used by Chen [39] and Norman [40] to consider the heating effect of the metal vapor plume on the keyhole.

$$k \frac{\partial T}{\partial \mathbf{n}} = q_l + q_{plume} - h_c(T - T_{ref}) - \varepsilon_r k_b(T^4 - T_{ref}^4) - 0.82 \frac{ML_v}{\sqrt{2\pi MRT}} P_0 \exp\left(\frac{ML_v(T - T_v)}{RTT_v}\right) \quad (12)$$

Where h_c and ε are convective coefficient and radiative emissivity, R represents ideal gas constant, and T_v denotes evaporation temperature.

Thermal boundary conditions for other surfaces:

$$k \frac{\partial T}{\partial \vec{n}} = -h_c(T - T_{ref}) - \varepsilon_r k_b (T^4 - T_{ref}^4) \quad (13)$$

Momentum boundary conditions:

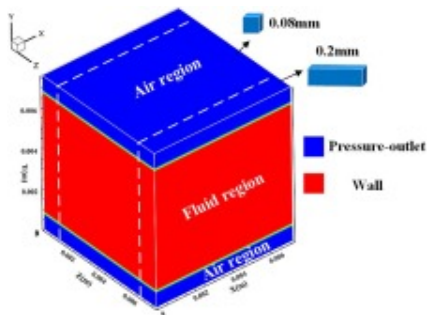
$$-P + 2\mu \frac{\partial v_n}{\partial n} = -P_r - P_{sta} + \frac{\sigma}{\kappa} \quad (14)$$

$$-\mu \frac{\partial v_t}{\partial n} = \frac{\partial \sigma}{\partial t} \frac{\partial T}{\partial S} + \tau_{vap} \quad (15)$$

Where $\mathbf{v}_n$ and $\mathbf{v}_t$ represent the normal and tangential velocity vectors, respectively. S indicates the radius of curvature of the free surface.

3.5. Model implementation

As shown in Fig. 3, the simulation domain used for the computation is divided into three subdomains: metal, upper gas-phase, and lower gas-phase domains. The overall size is 7.2mm×7.6mm×7.2mm. A uniform cubic ultrafine grid of a thickness of 0.08mm is utilized in both the welding direction and the weld depth direction. In the weld width direction, two distinct grid sizes are used: coarse and fine. A fine grid of 0.08mm is employed within a width of 5.2mm, while a coarse grid of 0.2mm is used to delineate the heat dissipation area. It ensures a sufficient area for a reliable measurement of the weld width while also significantly improving computational efficiency. The total number of meshes is 641,250. The boundary of the gas-phase regions is designated as pressure-outlets, whereas the metal domain is regarded as the wall boundary, considering thermal convection and radiation. The material thermophysical property values required for simulation are listed in Table 4 [41,42], and the temperature-dependent thermal conductivity and specific heat capacity are depicted in Fig. 4. All of the aforementioned control equations were discretely computed based on FLUENT 2019 R3. The simulation cases were run on a 32-core computer equipped with an AMD EPYC 7542 CPU and 256GB RAM.



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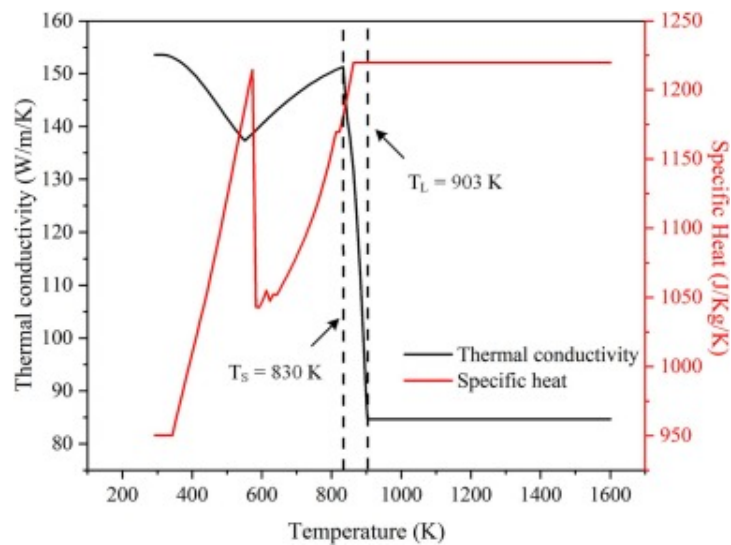
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Fig. 3. Computational domain required for the simulation (the white dashed lines represent the fine mesh region).

Table 4. Thermophysical property parameters of the 5A06 aluminum alloy.

Properties	Symbol	Unit	Value
Density	ρ	Kg/m ³	2630
Dynamic viscosity	μ	Kg/m/s	0.0013
Solidus temperature	T_S	K	830
Liquidus temperature	T_L	K	903
Boiling temperature	T_v	K	2700
Thermal conductivity	λ	W/m/K	Fig. 4
Specific heat	C_p	J/Kg/K	Fig. 4
Surface tension	σ	N/m	0.914
Surface tension gradient	$\partial\sigma/\partial T$	N/m/K	-0.00035
Latent heat of melting	H_{sl}	J/Kg	3.87×10^5

Properties	Symbol	Unit	Value
Latent heat of evaporation	L_v	J/Kg	1.08×10^7
Coefficient of thermal expansion	β	1/K	1.92×10^{-5}
Convective heat transfer coefficient	h_c	W/K/m ²	15
Thermal emissivity	ε		0.08



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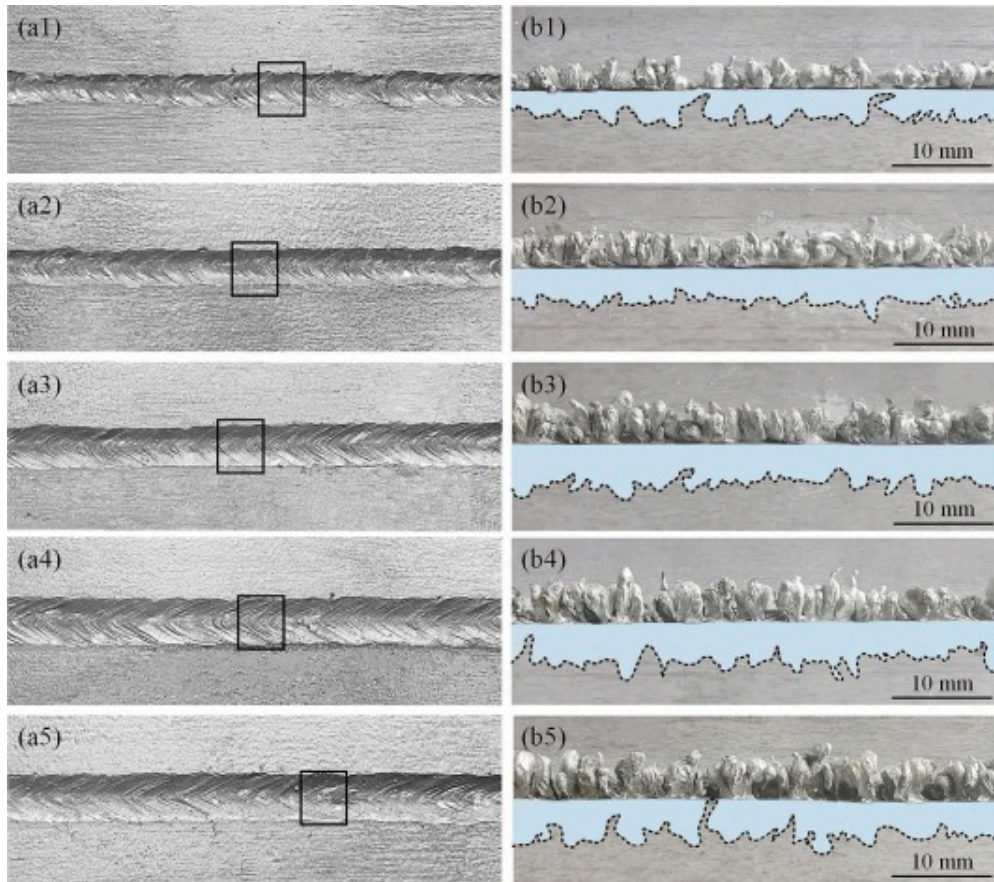
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Fig. 4. Temperature-dependent changes in specific heat and thermal conductivity (the black dashed lines represent the solidus and liquidus).

4. Results and discussion

4.1. Influence of ring beam on weld bead formation

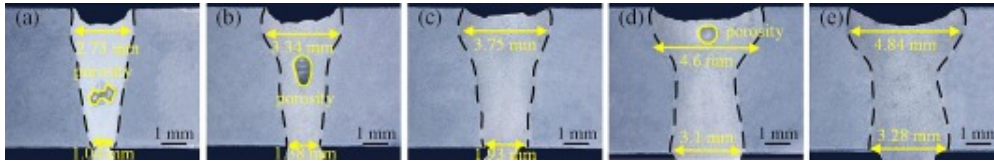
The surface and root morphology of full-penetration welds under different ring powers are depicted in Fig. 5(a1)-(b5). The weld width in the P_{C4} process was narrower and unevenly distributed along the welding direction, with a more pronounced undercut. A larger amount of spatter was attached to the weld surface, and the root was discontinuous, as shown in Fig. 5(a1) and (b1). The weld surface flatness improved significantly, the weld texture became more uniform, and the amount of spatter adhering to the surface decreased when the ring power was added and increased. At the same time, a more continuous weld root was formed, as illustrated in Fig. 5(a2)-(b4). Nevertheless, too high ring beam strength led to excessive energy input. The violent molten pool fluctuation led to an increase in spatter, and the increased penetration area also caused more molten metal to descend, resulting in a worse continuity of the weld root, as shown in Figs. 5(a5) and (b5). To demonstrate more intuitively the impact of varied ring powers on weld morphology and size, several weld cross-sections under the corresponding process were chosen as exhibited in Fig. 6, and the weld sizes were statistically as illustrated in Fig. 7. Fig. 6 and Fig. 7 showed that the weld seam produced by the P_{C4} process was smaller and tapered, with a significant collapse of the weld surface. Irregular porosity defects were found in the bottom and middle of the weld. As the ring power increased, the total heat input increased significantly, leading to greater metal melting. Under the influence of gravity, the molten metal fell, carrying more heat to the bottom of the weld and increasing the area of melting at the bottom. After the weld solidified, the bottom width of the weld approached the surface width, and the overall shape of the weld changed from a conical shape to a bolt shape, eventually presenting an "X" shape with little difference between the top and bottom widths. Appropriate ring power helped to relieve the collapse of the weld surface, but excessive further aggravated it. In addition, increasing ring power caused the pores to shift from the bottom and middle to the top of the weld, resulting in a decrease in porosity size and a more uniform distribution of pores.



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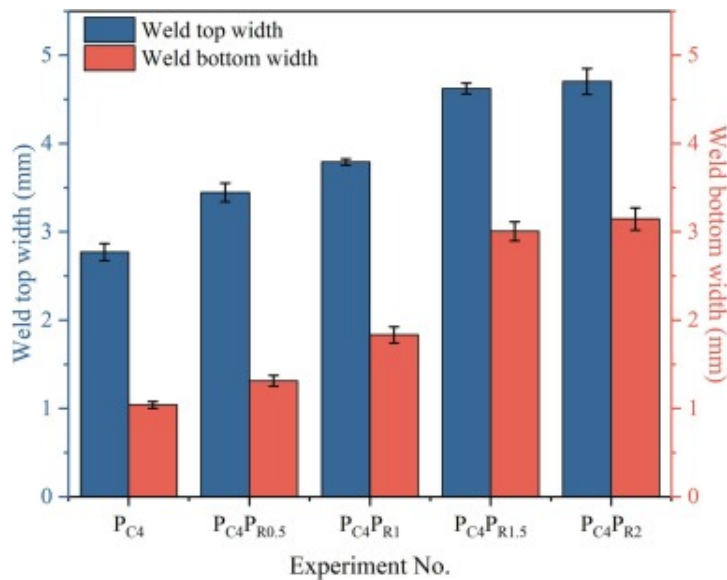
Fig. 5. Weld surface and root formation (the black dashed lines represent the weld root profile; the surface roughness test region is shown by the black rectangle): (a1-b1) ring power of 0W (P_{C4}); (a2-b2) ring power of 500W ($P_{C4}P_{R0.5}$); (a3-b3) ring power of 1000W ($P_{C4}P_{R1}$); (a4-b4) ring power of 1500W ($P_{C4}P_{R1.5}$); (a5-b5) ring power of 2000W ($P_{C4}P_{R2}$).



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Fig. 6. Cross-sectional morphology of weld (the black dashed lines represent the weld boundary; the yellow solid lines mark the pore defect): (a) ring power of 0W (P_{C4}); (b) ring power of 500W ($P_{C4P_{R0.5}}$); (c) ring power of 1000W ($P_{C4P_{R1}}$); (d) ring power of 1500W ($P_{C4P_{R1.5}}$); (e) ring power of 2000W ($P_{C4P_{R2}}$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

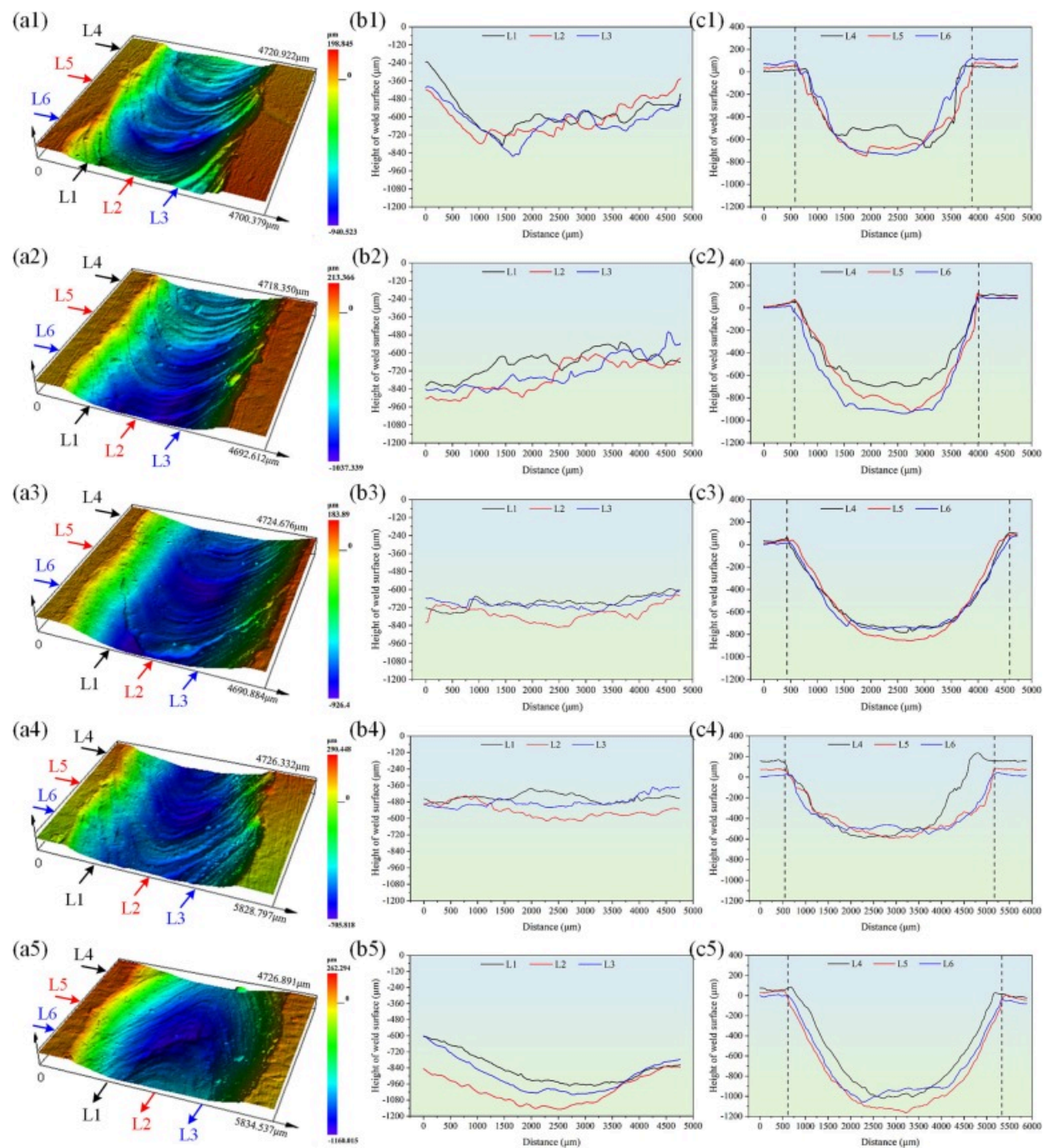


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Fig. 7. Width statistics of the top and bottom of the weld.

To further quantify the influence of the ring power on surface flatness, samples of the weld under various processes were extracted, as indicated in the black rectangular region of Fig. 5. Fig. 8 depicts the weld profiles and line-to-surface height at different ring powers. Lines L1, L2, and L3 indicates the profiles at various positions along the weld direction (Fig. 8(b1)-(b5)), while L4, L5, and L6 represents the profiles at different positions perpendicular to the weld direction (Fig. 8(c1)-(c5)). The range between the black dashed lines could determine the width of the weld. Table 5 lists the maximum line-surface height differences at different positions of the weld under various procedures, which can be used to measure surface homogeneity and collapse height. From the weld formation cloud maps in Fig. 8(a1)-(a5), it could be observed that with increasing ring power, the spatter on the weld surface initially decreased and then increased, while the weld continuity was greatly enhanced. Combined with Fig. 8(b1)-(b5), Fig. 8(c1)-(c5) and Table 5, the average maximum surface height differences along the weld direction under various ring powers were $490.31\mu\text{m}$ (P_{C4}), $344.87\mu\text{m}$ ($P_{C4}P_{R0.5}$), $184.27\mu\text{m}$ ($P_{C4}P_{R1}$), $163.10\mu\text{m}$ ($P_{C4}P_{R1.5}$), and $380.12\mu\text{m}$ ($P_{C4}P_{R2}$), respectively. This means that the ring beam made the weld surface more uniform and flatter, with the effect being most pronounced at 1500W ring power, resulting in a 66.74% reduction in the average maximum surface height difference compared with the P_{C4} . In addition, the average collapse heights of the vertical weld surfaces were $820.36\mu\text{m}$ (P_{C4}), $972.32\mu\text{m}$ ($P_{C4}P_{R0.5}$), $895.62\mu\text{m}$ ($P_{C4}P_{R1}$), $705.16\mu\text{m}$ ($P_{C4}P_{R1.5}$), and $1136.39\mu\text{m}$ ($P_{C4}P_{R2}$), respectively. It demonstrates that increasing ring power initially leads to worsening surface collapse, but later considerably improves. The least surface collapse height, 14.07% lower than that of the P_{C4} , was also found in the 1500W ring power. However, weld flatness and surface collapse were significantly harmed by further increasing the ring power.



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Fig. 8. Cloud map of surface formation and height statistics along and perpendicular to the weld direction: (a1)-(c1) ring power of 0W (P_{C4}); (a2)-(c2) ring power of 500W ($P_{C4}P_{R0.5}$); (a3)-(c3) ring power of 1000W ($P_{C4}P_{R1}$); (a4)-(c4) ring power of 1500W ($P_{C4}P_{R1.5}$); (a5)-(c5) ring power of 2000W ($P_{C4}P_{R2}$).

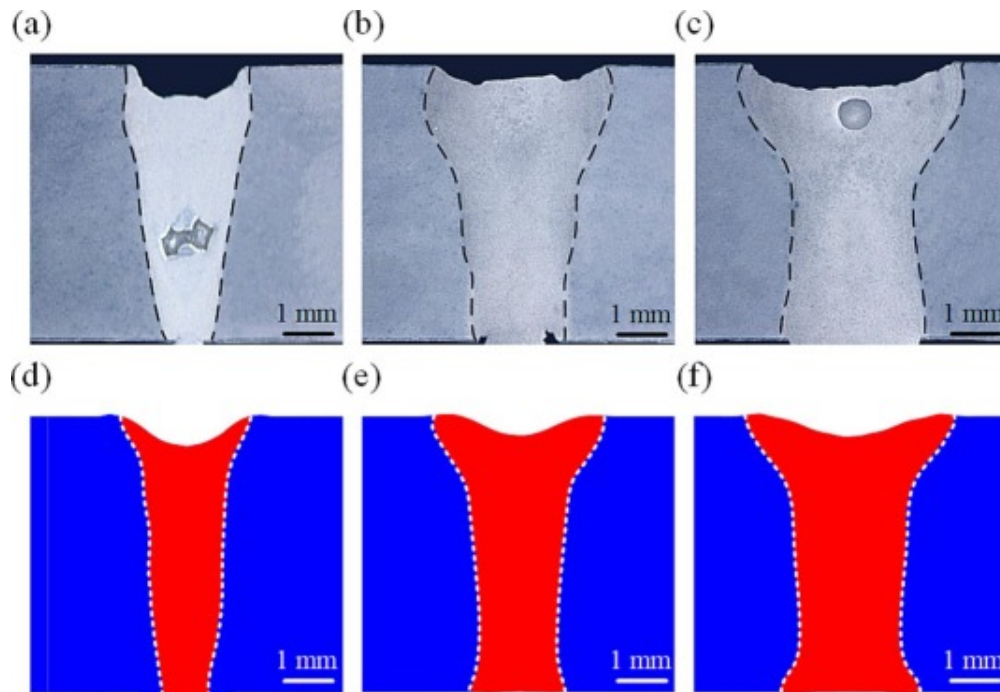
Table 5. Maximum surface height difference at different positions of the weld (unit: μm).

Location	P_{C4}	$P_{C4}P_{R0.5}$	$P_{C4}P_{R1}$	$P_{C4}P_{R1.5}$	$P_{C4}P_{R2}$
L1	564.23	305.59	172.43	126.99	374.99
L2	435.85	320.04	223.33	188.50	325.87
L3	470.85	408.97	157.10	173.81	439.49
Avg.	490.31	344.87	184.27	163.10	380.12
L4	735.60	824.88	894.47	828.41	1110.68
L5	834.34	1049.06	954.13	685.84	1224.53
L6	873.14	1043.82	838.27	601.22	1073.97
Avg.	820.36	972.32	895.62	705.16	1136.39

4.2. Model validation

The cross-sectional morphologies of the welds derived from the experiment (Fig. 9(a)-(c)) and simulation calculations (the red area in Fig. 9(d)-(e)) were compared, and the cross-sectional dimensions at different positions in the direction of penetration were statistically analyzed to verify the accuracy of the numerical calculation. Fig. 10 compares experimental and simulation results for the weld surface width (WSW), weld waist width (WWW), weld bottom width (WBW) and surface collapse height (SCH) in three typical

cases, indicating good agreement. The simulation framework is reliable in restoring typical physical processes and maintaining mathematical accuracy, as evidenced by the error ranges of 1.30% to 3.96%, 0.80% to 9.43%, 0.20% to 3.85% and 1.56% to 6.49% for WSW, WWW, WBW and SCH, respectively, under various ring powers. The existence of errors can be attributed to the difficulty of obtaining the real thermophysical parameters of materials at high temperatures in simulation and the simplification of physical processes, such as failing to account for the influence of shielding gas and ignoring the heating effect of plasma on the keyhole wall.

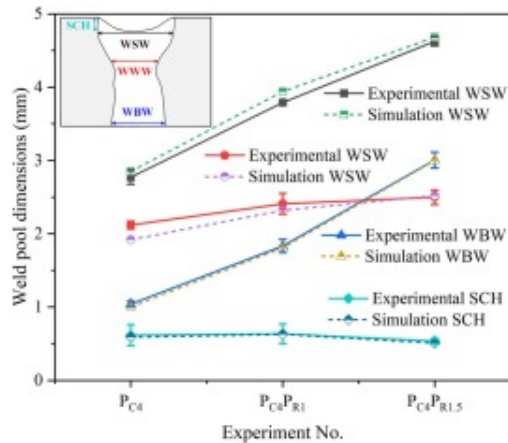


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Fig. 9. Comparison of experimental and simulated weld profiles (the black/white dashed lines are the weld boundary obtained from experimental and simulation, respectively). (a) and (d) are the experimental and simulation results for ring power of 0W (P_{C4}), respectively. (b) and (e) are the experimental and simulation results for ring power of

1000W ($P_{C4}P_{R1}$), respectively. (c) and (f) are the experimental and simulation results for ring power of 1500W ($P_{C4}P_{R1.5}$), respectively.

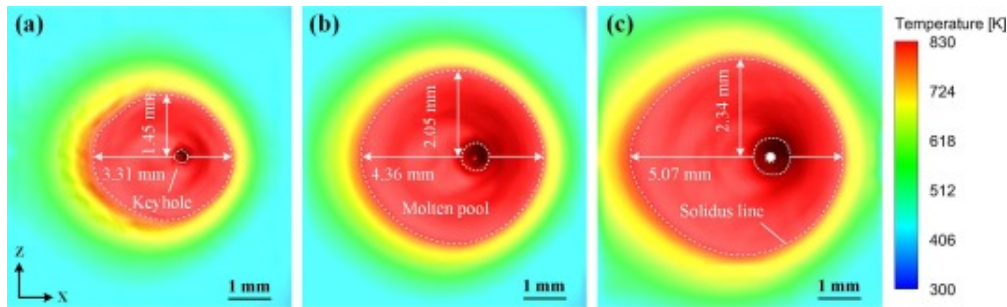


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Fig. 10. Comparison of experimental and simulation findings for weld surface width, waist width, bottom width and surface collapse height with various ring powers.

Fig. 11 depicts the simulation results for the molten pool morphology on the substrate's top surface (XZ plane) with various ring powers. The ring beam significantly expanded the keyhole entrance and molten pool area, which were positively correlated with ring power. The molten pool under the P_{C4} process was only 3.31 mm in length and 1.45 mm wide, respectively. The length of the molten pool in the $P_{C4}P_{R1}$ and $P_{C4}P_{R1.5}$ developed to 4.36 mm and 5.07 mm, respectively, while the width extended to 2.09 mm and 2.34 mm, illustrating the importance of the ring beam on the molten pool profile.



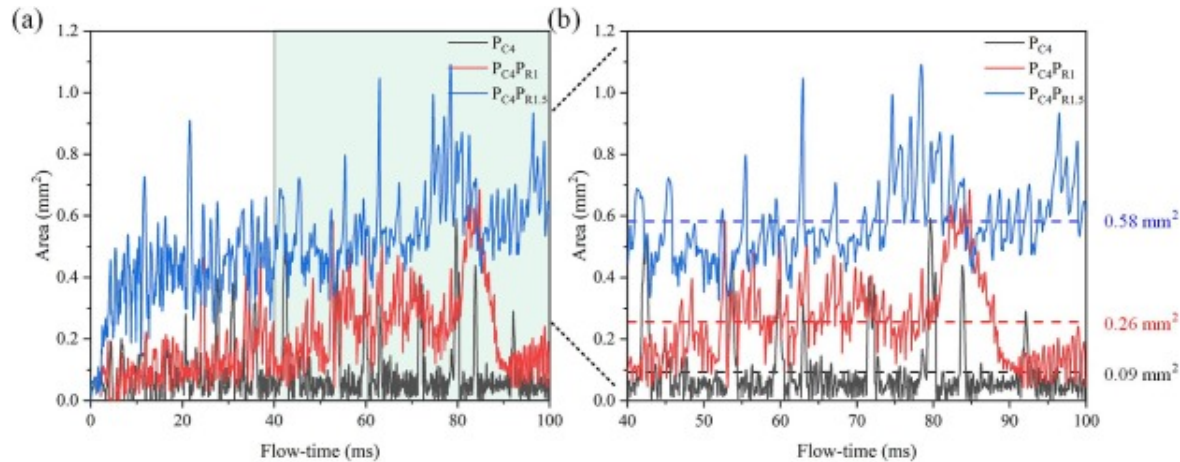
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Fig. 11. Simulation results of the top surface of the molten pool with various ring powers (the white dashed lines represent the border of keyhole entrance and the molten pool, respectively): (a) ring power of 0W (P_{C4}); (b) ring power of 1000W ($P_{C4}P_{R1}$); (c) ring power of 1500W ($P_{C4}P_{R1.5}$).

4.3. Influence of ring beam on keyhole dynamic behavior

Fig. 12 displays the fluctuation of the keyhole entrance area under various ring power levels. The oscillation curve of the keyhole entrance area from 40 to 100ms is indicated in Fig. 12(b). Only 28.97ms, or 48.28% of the time, was required for the keyhole entrance area to increase under the P_{C4} . The increase in ring power prolonged the time for the keyhole entrance area to rise by 35.49ms ($P_{C4}P_{R1}$), accounting for 59.15%, and 36.20ms ($P_{C4}P_{R1.5}$), accounting for 60.33%, respectively. This demonstrates that, under the action of the ring beam, the keyhole entrance remains open for a more extended period, thereby reducing keyhole closures. At the same time, the average entrance area of the keyhole improved from 0.09mm^2 (P_{C4}) to 0.26mm^2 ($P_{C4}P_{R1}$) and 0.58mm^2 ($P_{C4}P_{R1.5}$), respectively. The increase in the keyhole entrance area is attributed to the broader spectrum of heat input.

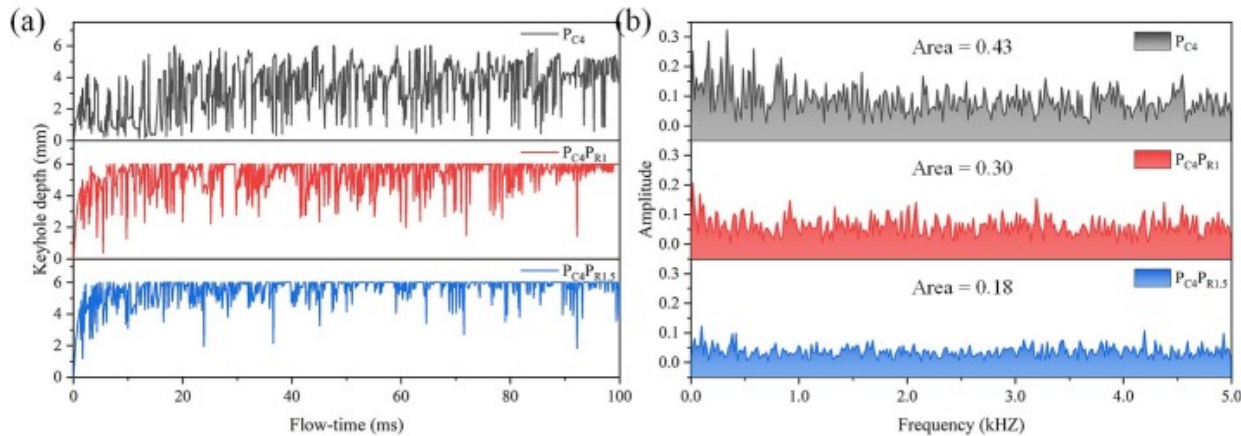


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Fig. 12. (a) changes in keyhole entrance area under different ring powers; (b) oscillation curve of keyhole entrance area in quasi-steady-state process (40–100ms).

Fig. 13 displays the keyhole depth oscillation curves and their associated Fourier space transformations for various ring powers. The keyhole fluctuates violently under the P_{C4} , and the keyhole collapsed, resulting in frequent entrance closures [43]. The ring beam significantly improved the oscillation behavior of the keyhole, dramatically reducing both the oscillation frequency and amplitude, as shown in Fig. 13(a). Furthermore, as Fig. 13(b) illustrates, the ring beam significantly reduced the area enclosed by the Fourier transform curve from 0.43 (P_{C4}) to 0.30 ($P_{C4}P_{R1}$) and 0.18 ($P_{C4}P_{R1.5}$), and the peak intensity in the Fourier space was weakened, indicating greater keyhole stability. This is because a broader and more stable keyhole entrance facilitated the escape of metal vapor, and the bottom of the keyhole absorbs laser energy, thereby enhancing the keyhole's ability to withstand molten pool contact and effectively preventing frequent collapse of the keyhole, which results in a more stable keyhole.



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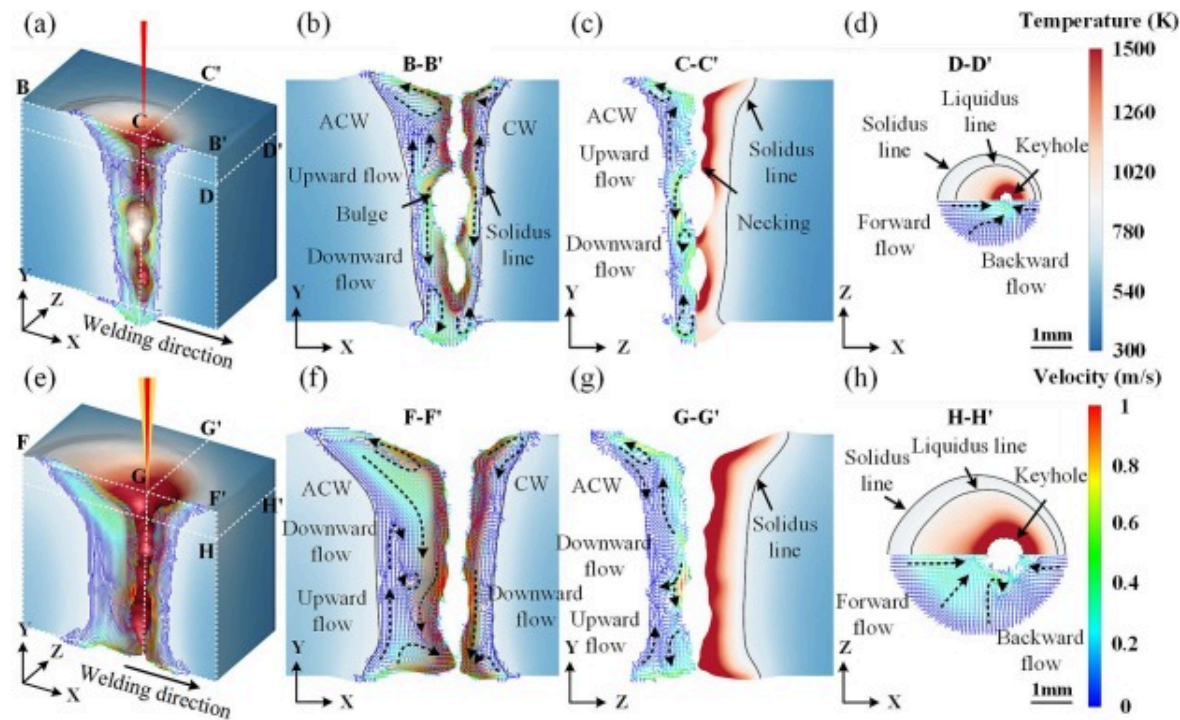
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Fig. 13. Keyhole depth fluctuation with different ring powers. (a) keyhole depth fluctuation curves; (b) Fourier transform of depth fluctuation.

4.4. Influence of ring beam on thermal-fluid characteristics

Fig. 14 depicts the temperature distribution and melt flow of a chosen cross-section of the molten pool during the P_{C4} and $P_{C4}P_{R1.5}$ welding processes. Fig. 14(a)-(d) displays the simulation results of the incomplete penetration moment of the P_{C4} , showing the typical melt flow distribution under partial penetration conditions [44]. Fig. 14(b) and (c) demonstrate that the higher melt flow rate is mainly concentrated near the surface and bottom of the molten pool, as well as the constriction position of the irregular keyhole and the bottom. The near-surface of the molten pool exhibited a typical surface tension-driven Marangoni convection. The melt fell due to its own gravity and recoil pressure, resulting in a downward flow at the bottom of the molten pool and a flow driven by the bottom surface tension. The laser beam with a high energy density caused intensive evaporation at the keyhole contraction in the middle of the molten pool. The recoil pressure drove the local metal liquid to flow downhill along the keyhole bulge, where it intersected with the upward-flowing metal liquid at the bottom of the keyhole and

impacted the bulge, resulting in contraction and subsequent collapse. The collision of multiple flows in the middle of the cross-sectional molten pool generated multiple flows and local vortices. As illustrated in Fig. 14(d), the melt flow in the horizontal section was relatively regular, mainly traveling from the molten pool's border to the keyhole wall, impacting the keyhole and forcing it to close.



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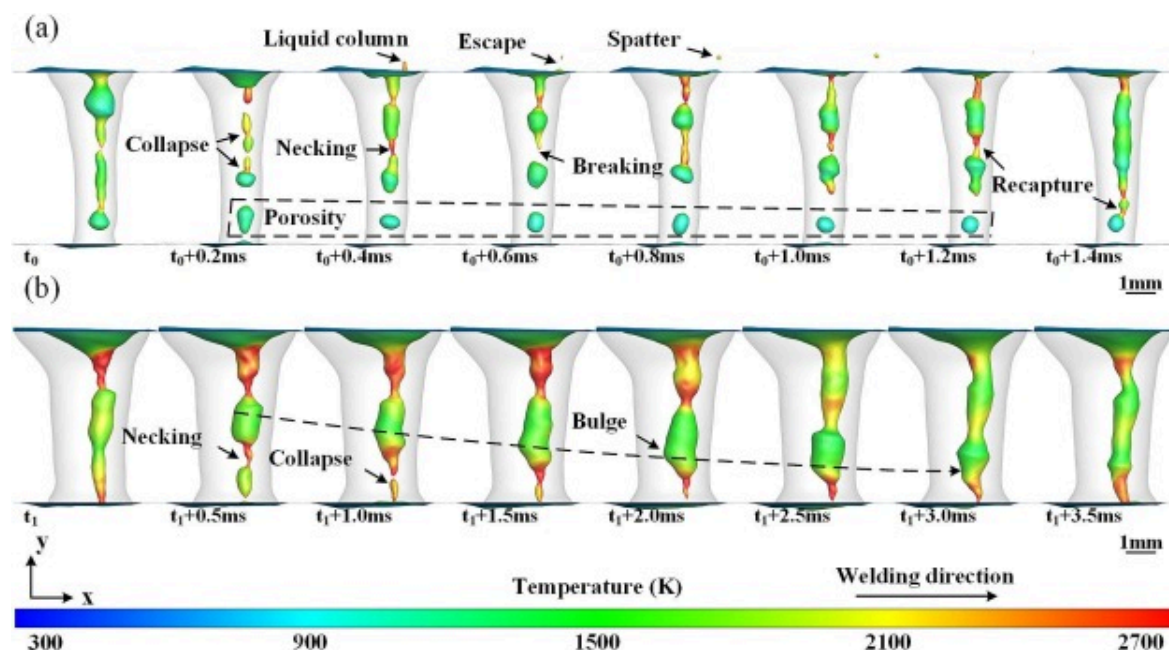
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Fig. 14. Temperature distribution and melt flow state in different typical cross sections (the white/black dashed lines represent the cross sections at different selected positions and keyhole boundary, respectively, and the black solid lines represent the solidus and liquidus, respectively). (a)–(d) are the three-dimensional distribution, longitudinal cross-section, cross section and horizontal plane with ring power of 0W (P_{C4}), respectively; (e)–(h) are the three-dimensional distribution, longitudinal cross-section, cross section and horizontal plane with ring power of 1500W ($P_{C4P_{R1.5}}$), respectively.

The molten pool of $P_{C4}P_{R1.5}$ exhibits a broader range of temperature and flow distribution, as shown in Fig. 14(e)-(h), which displays the simulation results concurrently with P_{C4} . It was manifested as a fully penetrated through-keyhole state with a smoother and more regular keyhole wall. The melt flow was significantly different from that of P_{C4} , and the overall performance was multiple long-distance regular flows. The front wall of the keyhole primarily exhibited a top-down flow and a local anticlockwise (ACW) vortex at the bottom. The Marangoni circulation on the upper surface was obstructed by the solidification front and turned downhill, becoming the primary flow near the keyhole surface in the middle of the molten pool. In addition, when the through keyhole was formed, the recoil pressure and vapor shear force led the bottom melt to flow backwards. After being blocked by the solidification front, it turned upward and collided with the downward flow, generating multiple small local vortices in the middle and a clockwise (CW) vortex at the bottom, which, together with the ACW vortex at the bottom of the front end, impacting the keyhole and causing it to close, as illustrated in Fig. 14(f). Fig. 14(g) illustrates that in the melt flow behavior of the cross section. It exhibited minor variations in the middle of the keyhole, owing mostly to recoil pressure, which generated two flows, upward and downward, at the small bulge. The upward flow merged into the upper surface circulation, and the downward flow impacted the bottom of the keyhole. Furthermore, an additional reverse flow was observed through the keyhole wall toward the borders of the molten pool in the horizontal cross-section under the $P_{C4}P_{R1.5}$, as shown in Fig. 14(h), which can effectively alleviate the force of the metal on the keyhole and maintain a larger and more stable keyhole entrance.

Fig. 15 illustrates the keyhole oscillation and wall temperature distribution under the conditions of P_{C4} and $P_{C4}P_{R1.5}$, with the gray region indicating the border of the molten pool. The chosen time period is the typical cycle from the oscillatory fracture of the keyhole to the recapture of the regular keyhole. There were multiple bulges on the keyhole at the initial time (t_0) in the P_{C4} , and the collapse was severe at $t_0+0.2$ ms. The bulges on the upper part of the keyhole traveled upward, while the bottom bulges fractured to generate pores and collapsed in stages in the middle. The collapse caused a

local high-temperature zone to form on the top section of the keyhole. Strong metal eruption struck the irregular keyhole wall. The molten pool expanded and entrained under vapor shear, entraining the upward-moving metal liquid to produce a liquid column. Meanwhile, the recoil pressure prompted the keyhole to reopen and reconnect with the collapsed keyhole part in the middle, as shown at $t_0+0.4\text{ms}$. Due to the shear force of the metal vapor and its own gravity, the liquid column necked and separated into droplets, producing spatters. The separated liquid column dropped down into the molten pool, striking its surface and causing it to fluctuate violently. A similar phenomenon was also found in the experimental observations of Zhang [45]. Simultaneously, this violent fluctuation shielded the laser energy, preventing it from reaching the middle of the keyhole stably, which in turn caused it to repeatedly break and connect, such as at $t_0+0.6\text{ms}$ – 1.2ms . After the spatter had escaped, the fluctuation of the molten pool surface was alleviated, and the keyhole was reopened, allowing the laser energy to reach the bottom of the keyhole. This further evaporated the molten metal between the keyhole and the bubble, and ultimately recaptured the bubble, as shown at $t_0+1.6\text{ms}$. During this process, the bubbles at the bottom approached a spherical shape due to the surface tension and eventually reached their lowest surface energy [46]. Such “lucky moments” are rare. Usually, bubbles that are too large will fall into the bottom molten pool due to gravity, while the solidification front will trap bubbles that are too small to form pores.



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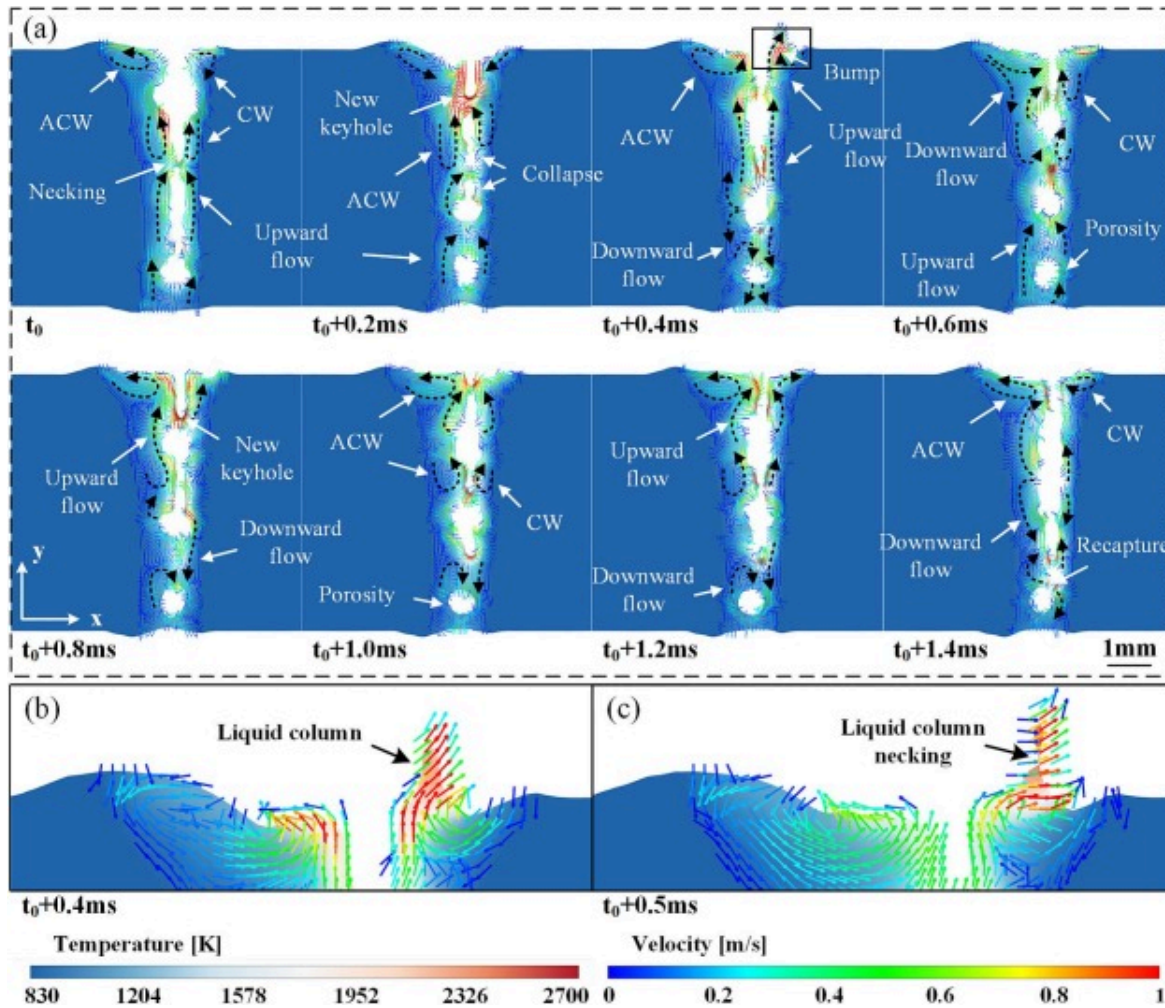
Fig. 15. Oscillation behavior of the keyhole and wall temperature distribution in a typical period (the gray region indicates the molten pool). (a) ring power of 0W (P_{C4}); (b) ring power of 1500W ($P_{C4P_{R1.5}}$).

Fig. 15(b) shows that the $P_{C4P_{R1.5}}$ provides a larger and more stable keyhole entrance and channel than the P_{C4} . At the initial time t_1 , a through keyhole was generated. The bulge in the middle of the keyhole concentrated the energy at the top of the bulge. More laser energy was shielded, causing the bottom of the bulge to shrink and collapse, as depicted at $t_1 + 1.0$ ms. However, the wider keyhole entrance and smoother keyhole wall reduced melt shearing by melt vapor, preventing molten pool expansion and the development of liquid columns, resulting in a flatter molten pool surface. Additionally, the wider keyhole entrance enhanced the ability to drill down and recapture the collapsed keyhole, such as $t_1 + 1.5$ ms. The laser beam irradiated the top of the bulge, forming an incredibly dense

energy area that generated a greater recoil pressure, which only expanded the keyhole channel, such as $t_1+2.0\text{ms}$ and $t_1+2.5\text{ms}$, but also pushed the bulge down to reach the keyhole bottom, ultimately reforming a through keyhole, such as $t_1+3.0\text{ms}$ and $t_1+3.5\text{ms}$. The downward movement of the protrusion has a very positive effect on capturing bubbles and resisting keyhole collapse. Additionally, compared with the keyhole oscillation period of 1.4ms for the P_{C4} , the period of 3.5ms for the $P_{C4}P_{R1.5}$ also demonstrates a more stable molten pool and keyhole, which substantially reduces the frequency of spatter generation and obtains a flat molten pool surface.

The transient behavior of the keyhole exhibits substantial periodic characteristics [36], namely the repeated drilling and keyhole collapse process. The keyhole is subjected to recoil pressure, impact, and vibration of metal liquid, leading to defects such as spatter, surface collapse, and porosity. Fig. 16 describes the keyhole oscillation behavior and melt heat-flow distribution in the longitudinal section under the P_{C4} . At time t_0 , the solidification front hindered the ACW near the surface and turned it downward to merge with the upward flow at the bottom, producing continuous harm to the middle of the keyhole. Severe multi-segment collapse led to the formation of a new keyhole, as well as multiple independent small spaces and bubbles, such as those observed at $t_0+0.2\text{ms}$. The newborn keyhole preferentially absorbed laser energy to form a high-energy-density area. The intense melt evaporation reopened the keyhole channel, generating a high-speed flow upward along the keyhole wall. In addition, as illustrated in Fig. 16(b), the ACW on the rear surface and the CW on the front surface affected both sides of the top keyhole, respectively, creating a liquid column in front of the keyhole and a local bulge at the rear surface. The liquid column rose due to vapor shear and the impact of CW of the front keyhole wall. When it reached a particular height, it was separated from the droplets by its own gravity and the normal force of the vapor, and spatters were generated. The bulge on the rear surface merged back into the molten pool due to ACW and its own gravity, resulting in a flat rear surface, as illustrated in Fig. 16(c). Because of the reopening of the keyhole channel, the laser irradiates the bulge and bottom of the keyhole, producing a rising flow close to the keyhole wall near the bulge, which also leads

to generate the liquid column. After separating the spatter, the liquid column fell and struck the surface, resulting in downward metal reflux, such as $t_0+0.6\text{ms}$. Multiple flows lead to the keyhole collapsing and reconnecting frequently, and high-energy zones frequently appear above the keyhole. The melt falls due to the combined forces of recoil pressure and gravity, resulting in surface collapse. The bubbles produced at $t_0+0.2\text{ms}$ were “polished” smoother by the periodic upward and downward flow of the bottom, and their shape approached a spherical shape at $t_0+1.2\text{ms}$. Direct irradiation of the laser beam at the bottom of the keyhole allowed bubbles to be recaptured by the keyhole, which would otherwise be arrested by the solidification front, thereby generating a pore, such as $t_0+1.4\text{ms}$.

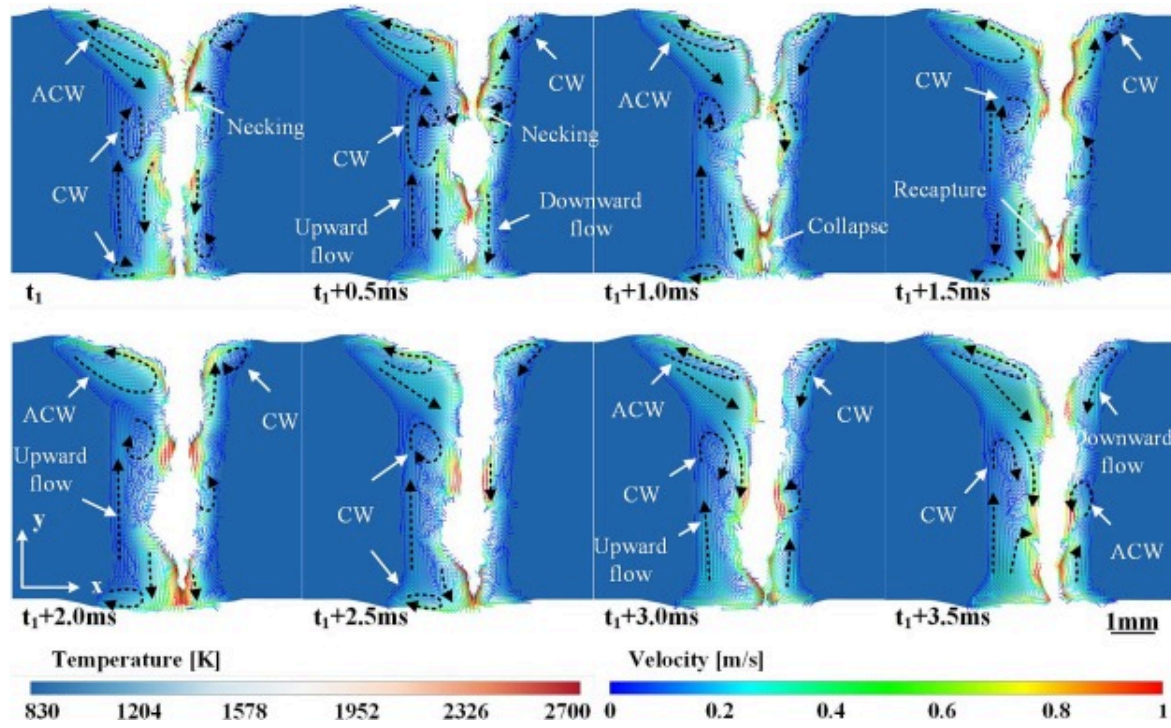


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Fig. 16. (a) Keyhole oscillation behavior and melt heat-flow distribution in a typical period in longitudinal section of the molten pool with ring power of 0W (P_{C4}); (b) the moment of liquid column formation; (c) the moment of liquid column shrinks and separates spatters.

As seen in Fig. 13, the $P_{C4}P_{R1.5}$ exhibited more pronounced FPLW characteristics: intermittent through keyhole formation. The keyhole oscillation behavior and melt heat-flow distribution in the longitudinal section under the $P_{C4}P_{R1.5}$ are shown in Fig. 17. A typical cycle from creating a through keyhole to reconstructing one was also chosen, and a similar ACW existed near the surface. At the initial moment (t_1), more laser energy was accumulated on the bulge surface of the upper keyhole. At the same time, recoil pressure pushed the melt to flow downward across the keyhole wall to the bottom of the molten pool. After merging with the surface tension-driven flow, it traveled backwards. It was impeded by the solidification front and turned upward, merging with the downward flow to produce a local small CW eddy. Since the through keyhole enabled the laser beam to escape from the bottom, less laser energy decreased the recoil pressure at the bottom. The metal liquid struck on both sides, and surface tension worked together to close the keyhole and build a liquid bridge, as seen at $t_1+0.5\text{ms}$ and $t_1+1.0\text{ms}$. As the laser beam worked on the liquid bridge, the collapsed keyhole was recaptured to achieve further drilling, such as $t_1+1.5\text{ms}$. The bulge moved downward under the recoil pressure at its top and reached the bottom of the molten pool, reshaping it into a through keyhole, such as $t_1+3.5\text{ms}$. After forming the through keyhole, the shielding of the upper surface from laser energy is weakened, allowing more beams to reach the keyhole bottom, which in turn cause the bottom temperature to be higher than that of the upper surface. The bottom exhibits unstable shape fluctuations because of the lower surface tension. It has been discovered that the keyhole can maintain a wider channel throughout the entire cycle of the $P_{C4}P_{R1.5}$, ensuring sufficient recoil pressure at the constriction to resist the force of the melt. Meanwhile, the CW and ACW downward flow on both sides of the top keyhole inhibit the upward flow of the melt, effectively preventing the upward flow from expanding the molten pool and causing surface fluctuations. Even if there were a faster upward flow at individual moments, such as $t_1+1.5\text{ms}$ and $t_1+2.0\text{ms}$, creating minor local bulges, they would be reintegrated into the molten pool surface due to the rightward and downward flow. Furthermore, opening the keyhole more frequently helps to release the pressure and heat in the molten pool promptly, preventing the melt from falling or even being lost.



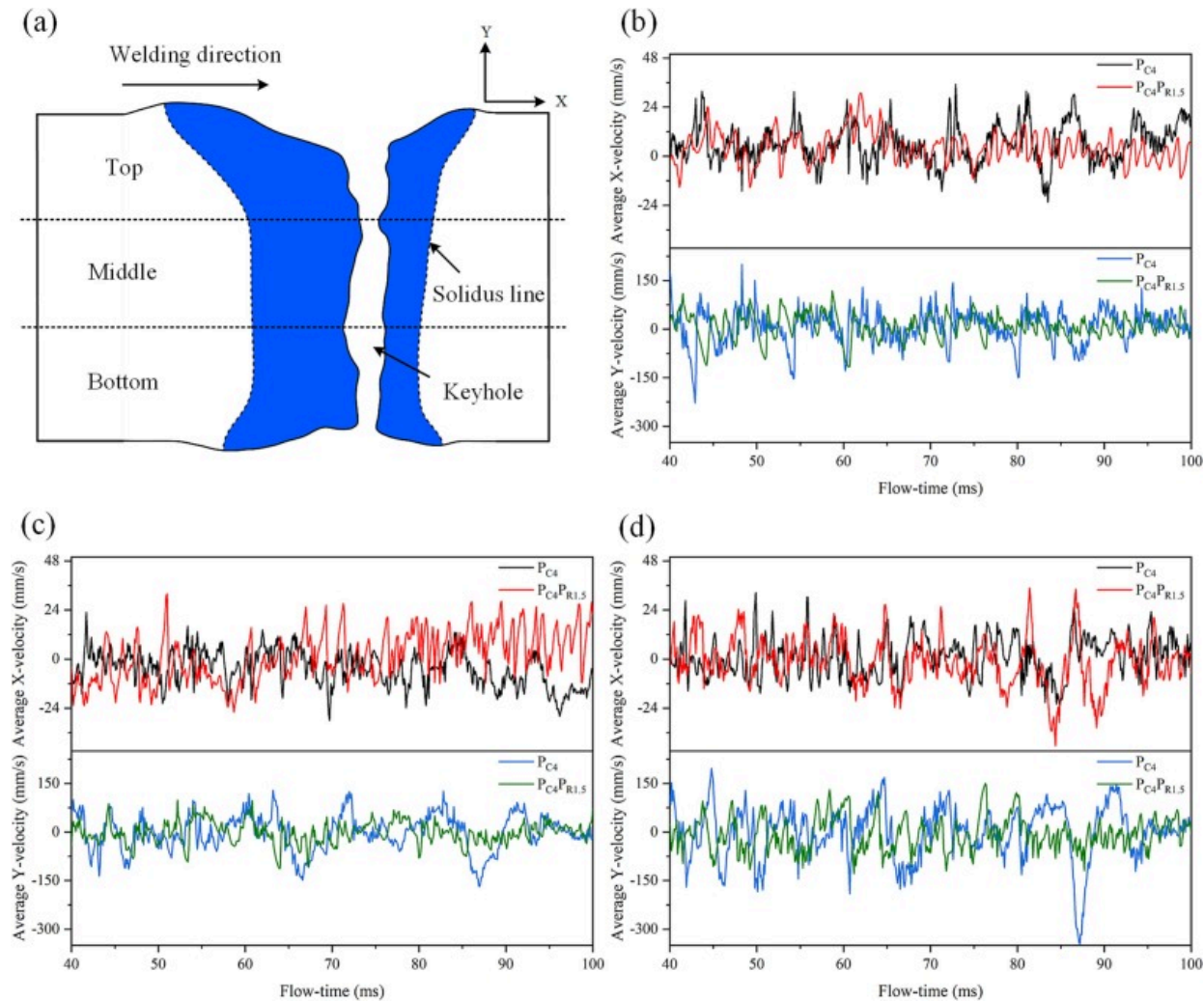
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Fig. 17. Keyhole oscillation behavior and melt heat-flow distribution in a typical period in longitudinal section of the molten pool with ring power of 1500W ($P_{C4}P_{R1.5}$).

The above described the molten pool flow behavior under both modes. To further quantify the fluid flow rate and analyze the flow velocity distribution in the welding direction (X) and the keyhole depth direction (Y) in three-dimensional space, thereby revealing the formation process of spatter and surface collapse, as well as the mechanism of flow mode to enhance them. As depicted in Fig. 18(a), based on the Y direction, the molten pool was separated into three regions: top, middle, and bottom. The fluid zone above the solidus (the blue area in Fig. 18(a)) was chosen to more accurately reflect the general characteristics of welding stability [47]. Fig. 18(b)-(d) displays the average velocity fluctuation curves in different sections of the molten pool, and the corresponding

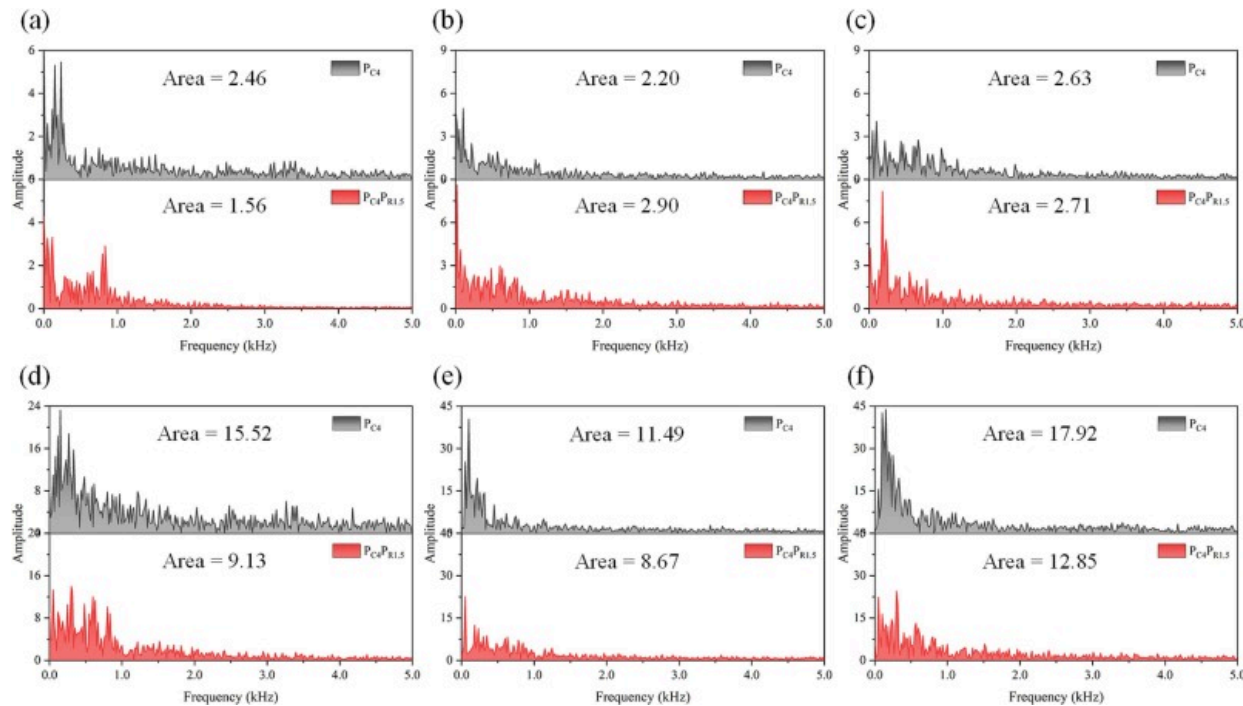
Fourier space transforms are shown in Fig. 19(a)–(f), which being used to describe the stability of the melt flow rate in different regions. Table 6 depicts statistical data on average velocity values in various directions over the molten pool. Combining Fig. 18 and Fig. 19, the average velocity amplitude in the Y direction and the top in the X direction, as well as the area of the Fourier transform curve envelope and the peak intensity, were considerably decreased with $P_{C4}P_{R1.5}$, indicating that the overall melt flow is more stable. In comparison to P_{C4} , at the top of the molten pool, the average flow velocities in the -X and -Y directions decreased by 29.03% and 37.25%, respectively. The flow velocities in the +X and +Y directions decreased by 20.47% and 20.23%, respectively, with the $P_{C4}P_{R1.5}$. Smaller +X and +Y directions' flow velocities, along with more stable melt flow, indicates a weaker influence on the keyhole and the capacity to build a liquid column. The top of the keyhole can maintain a stable entrance and effectively avoid splashing generated by the upward flow. The lower temperature gradient induced by the wider molten pool and the smooth keyhole wall are the primary reasons for the decrease of the -X and -Y directions' flow velocities. In the middle and bottom regions, due to the development of a more robust keyhole channel, the laser beam may frequently reach these areas. The downward movement of the energy will produce a larger average flow velocity in the X direction and a less stable melt flow, which becomes an important reason for the bottom keyhole to close and end the through keyhole, as seen in Fig. 18(c) and (d), Fig. 19(b) and (c), and Table 6. Since the laser energy under the P_{C4} frequently acted on the protrusion and liquid bridge, resulting in a higher average flow velocity in the Y direction, which mainly manifested as a downward flow in the -Y direction. In contrast, the $P_{C4}P_{R1.5}$ had a smaller -Y direction flow, and the average -Y direction flow velocity in the middle and bottom regions was reduced by 41.96% and 40.48%, respectively. The smaller and more stable -Y flow and the larger +Y flow (with an average flow velocity of 31.79 mm/s), which dominates the top, jointly supports the melt, effectively alleviating it from falling under the multiple effects of gravity and recoil pressure, thus decreasing the collapse height of the molten pool.



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Fig. 18. Average X and Y direction velocity distribution in various regions of the molten pool with ring power of 0W (P_{C4}) and 1500W ($P_{C4}P_{R1.5}$). (a) schematic diagram of the molten pool area division for statistical average velocity; velocity fluctuation curves for the (b) top regions; (c) middle regions; (d) bottom regions.



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Fig. 19. Fourier transform of the average flow velocity in various regions. (a)–(c) are the Fourier transforms of the average X flow velocity at the top, middle and bottom, respectively; (b)–(d) are the Fourier transforms of the average Y flow velocity at the top, middle and bottom, respectively.

Table 6. Average velocity in the X and Y directions of different regions of the molten pool.

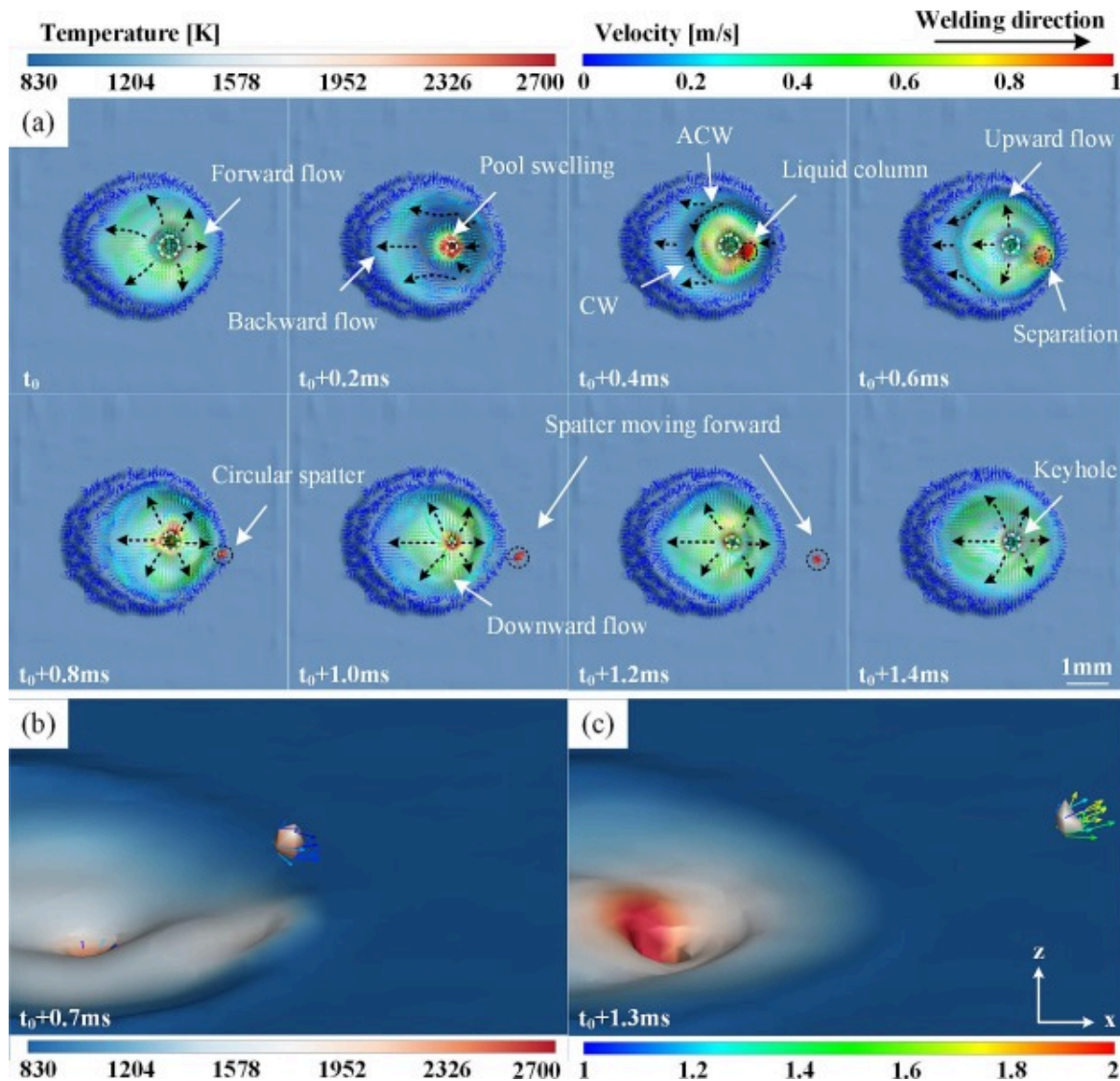
Area	Average Velocity (mm/s) of P_{C4}				Average Velocity (mm/s) of $P_{C4}P_{R1.5}$			
	+X	-X	+Y	-Y	+X	-X	+Y	-Y
Top	9.77	5.96	39.85	43.20	7.77	4.23	31.79	27.11
Middle	4.77	9.06	41.03	45.81	10.93	8.99	25.80	26.59

Area	Average Velocity (mm/s) of P_{C4}				Average Velocity (mm/s) of $P_{C4}P_{R1.5}$			
	+X	-X	+Y	-Y	+X	-X	+Y	-Y
Bottom	7.85	7.56	56.60	71.22	8.42	9.57	40.76	42.39

In addition, the $P_{C4}P_{R1.5}$ resulted in a larger -X flow velocity (9.57 mm/s) at the bottom of the molten pool, allowing bubbles that were not captured by the keyhole to move faster to the bottom rear, thereby increasing the possibility of bubble escape. In comparison, the bubbles under the P_{C4} suffered a “catastrophic blow”. The smaller -X flow velocity (7.56 mm/s) and the larger -Y flow velocity (71.22 mm/s) makes it difficult for the bubbles to escape, thereby increasing the probability of being trapped by the solidification front and contributing to pore formation.

The heat-flow distribution on the top surface of the molten pool of the P_{C4} is shown in Fig. 20, which visually shows the formation and escape process of spatter. It primarily manifested as flow from the keyhole entrance to the edge of the molten pool. Significant expansion and contraction of the keyhole entrance could be observed, indicating its instability. At time t_0 , the top surface of the molten pool exhibited a relatively uniform flow velocity distribution and transition from the keyhole to the periphery. The high-speed spraying of metal vapor produced a massive vertical upward flow when the keyhole collapsed and a new one formed, which further expanded the molten pool. The melt squeezed the keyhole entrance, causing it to shrink, as shown at $t_0+0.2$ ms. The metal liquid in front of the keyhole rose due to the vapor shear force. At times, such as $t_0+0.4$ ms, the molten pool swelled into a liquid column, producing noticeable wrinkles on the surface. In contrast to other moments, there was a flow from the border to the keyhole at these two moments, which posed a severe challenge to the stability of the keyhole entrance. As depicted in Fig. 20(b), after the liquid column separated the droplet and generated spatter, the spatter flew out at 1 m/s at a certain angle to the welding direction, accelerated under the action of the vapor plume, and finally escaped from the calculation domain at 1.8 m/s, as seen in Fig. 20(c). Compared with the P_{C4} , the top surface

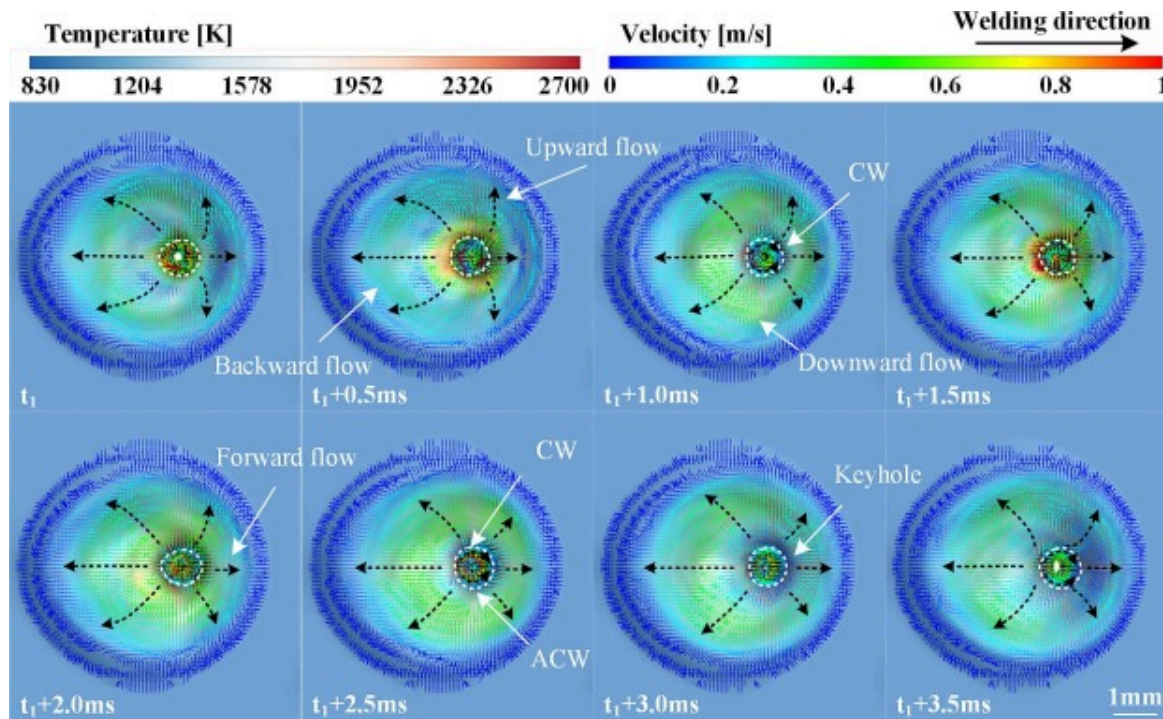
of the molten pool with the $P_{C4}P_{R1.5}$ exhibited a more stable keyhole entrance and a wider flow range, as shown in Fig. 21. At all times, the flow was uniform from the keyhole to the periphery, which prevented the keyhole from being struck by the melt. Additionally, compared with the molten pool bulges and surface wrinkles created by the P_{C4} , the more uniform flow velocity distribution and transition of the $P_{C4}P_{R1.5}$ alleviated its expansion and contraction, leading to a smoother molten pool surface. From another perspective, it is also confirmed that the ring beam can effectively promote surface formation.



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Fig. 20. (a) flow characteristics of top surface of the molten pool with ring power of 0W (P_{C4}) (white dashed line marks the keyhole boundary); (b) spatter acceleration stage; (c) the moment when spatter escapes from the computational domain.



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Fig. 21. Flow characteristics of top surface of the molten pool with ring power of 1500W ($P_{C4P_{R1.5}}$) (white dashed line marks the keyhole boundary).

5. Conclusion

This paper innovatively applied ARMLW to address the problems of weld spatter and surface collapse in the FPLW process of medium-thick aluminum alloy plates. The effect of the ring beam on the dynamic behavior of the keyhole and the flow characteristics of the molten pool during the FPLW of medium-thick aluminum alloy was investigated using a combination of experimental and numerical simulation methods. A verified 3D transient CFD model was employed to elucidate the mechanism by which the ring beam mitigated spatter and weld surface collapse during the FPLW. This study demonstrates

that applying ARMLW to address the issues of weld spatter and surface collapse during FPLW of medium-thick aluminum alloy is a reliable technical method. The main conclusions are summarized as follows:

1. The experimental findings demonstrated that the ring beam significantly enhanced the surface roughness and surface collapse height of the full-penetration weld, thereby achieving a uniform surface. The surface collapse height was reduced by 14.07% under 1500W ring power; however, further increased in ring power would have a negative impact.
2. By extending the average keyhole entrance area and prolonging the open time of the keyhole entrance, the ring beam considerably reduces keyhole oscillation and preserves a broader, more stable keyhole channel. Laser energy is radiated to the middle and bottom keyhole, resulting in a more even energy distribution on the keyhole wall.
3. In the typical keyhole cycle, the molten pool, under the influence of the ring beam, exhibited flow characteristics over a longer distance and a wider range. The average +X and +Y flow velocities at the top of the molten pool were 20.47% and 20.23% respectively. The average -X flow velocity was 29.03%, which effectively alleviated the shearing of metal vapor and the influence of the melt on the keyhole, as well as prevented the formation of the liquid column and spatter caused by the upward flow.
4. Under the action of the ring beam, the values of the average -Y velocity in the top, middle, and bottom molten pools were decreased by 37.25%, 41.96% and 40.48%, respectively. The lower downward flow velocity level and the dominant upward flow at the top effectively enhance the collapse of the weld surface.

This study enhanced the understanding of the mechanism of ring beam suppression on spatter and weld surface collapse in FPLW, providing valuable insights for the application of ARMLW technology in medium-thick plate aluminum alloy welding. During the study, we found that the ring beam causes the average flow velocity at the bottom of the keyhole to be larger in the -X direction and smaller in the -Y direction, indicating a potential improvement in the porosity defect. In the future work, we will reproduce the bubble generation process and its migration path to elucidate further the suppression mechanism of the ring beam on porosity defects. Simultaneously, we will utilize high-speed photography to monitor the welding process, thereby acquiring dynamic fluctuations of the surface molten pool and the transient behavior of the internal keyhole, to further verify the reliability of the created model framework.

CRediT authorship contribution statement

Kai Guo: Writing – original draft, Visualization, Methodology, Investigation. **Xinwei Du:** Validation, Methodology. **Meng Wang:** Resources. **Jicheng Chen:** Supervision, Resources, Conceptualization. **Yanhong Wei:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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